

Mounica Paladugu

meismounica079@gmail.com | 206-591-5835 | Mountain View, CA
[linkedin.com/in/mounica-paladugu](https://www.linkedin.com/in/mounica-paladugu) | [mounica-000.github.io](https://github.com/mounica-000)

HIGHLIGHT

Full-stack software engineer with hands-on experience in **backend systems**, **LLM-powered agent architecture**, and **distributed data pipelines** across production environments. Built and shipped systems ranging from a **RAG pipeline** with **89% retrieval precision** to a **6x database latency improvement**, with a track record of end-to-end ownership from architecture decisions to **scalable, deployed product**.

EDUCATION

Drexel University

Bachelor of Science in Computer Science, Minor in Mathematics

Philadelphia, PA

Sep 2020 – Jun 2025

WORK EXPERIENCE

Software Engineer

Develop For Good

Oct 2025 – Present

Remote, CA

- Led **end-to-end development** of a food recovery platform (**React, TypeScript, Python, FastAPI**); collaborated closely with stakeholders to define the **technical roadmap and product vision**, ensuring the platform met critical community needs within a three-month launch window.
- Designed a **headless CMS architecture** using **Airtable** as a dynamic data layer, eliminating client dependency on engineers for content updates and cutting change request turnaround from **days to minutes**; authored **system architecture documentation** for long-term maintainability.
- Drove post-launch growth to **25+ MAU** within two weeks by integrating **Google Analytics** to identify friction points and iterate on UX using **data-driven insights**.

Backend Software Engineer Intern

Rajant Corporation

Mar – Sep 2023

Malvern, PA

- Architected a **scalable** backend for **high-throughput telemetry metrics**, migrating to InfluxDB to reduce query latency ~6x (350–500ms → 50–90ms) and improving system **observability** for large-scale data ingestion.
- Designed a **RESTful API (Python/FastAPI, Docker)** with **10+ endpoints** supporting multi-resolution health data aggregations and enabling seamless integration with hospital records and the internal AI platform.
- Automated a real-time **data pipeline in Python** that watched a local directory and propagated changes to **MinIO S3 cloud storage** automatically, eliminating **5+ hours of manual work per week**.

Product Development Engineer Intern

Pression LLC

Mar – Sep 2022

Coatesville, PA

- Automated the test data pipeline in **Python**, reducing file storage costs by 25% and cutting data loss by 50%.
- Led the medical device transition from an unstable v1 prototype to a **reliable manufacturing-ready v2**, integrating **embedded C firmware** with system UI software, achieved **zero field failures** across all test scenarios, and enabled the company to pursue **FDA clearance filing**.
- Refactored and debugged **2000+ lines of broken UI code**; authored comprehensive **system architecture documentation** to ensure long-term maintainability and code quality.

PROJECTS

GitHub Issue Root Cause Agent | *Python, FastAPI, ChromaDB, Groq API, Streamlit UI*

Mar 2026

- Engineered an **LLM-powered agent architecture** with a **modular RAG pipeline**, utilizing ChromaDB for semantic retrieval and Llama 3.3 for automated root cause analysis.
- Developed a **semantic retrieval system** using **cosine similarity** to surface the top-3 most relevant past issues for any new bug report, achieving **89% retrieval precision** across paraphrased query variants in end-to-end evaluation.
- Integrated **Llama 3.3 70b** via **Groq API** with a structured prompt template grounded in retrieved context, returning actionable root cause analysis and fix suggestions; exposed as a **REST API** via **FastAPI** with a **Streamlit** demo UI.

Virtual Pet | *TypeScript, React, HTML/CSS, MySQL, Node.js*

Mar – May 2025

- Built a **full-stack real-time application** with **WebSocket**-based communication, **maintainable** normalized database schema, and **TypeScript** for end-to-end type safety.
- Deployed on a **Linux** (Linode) cloud server using **React, TypeScript, and MySQL**; wrote **unit and integration tests** to ensure code reliability.

SKILLS

Programming Languages: Python, Java, C, JavaScript/ TypeScript, SQL, Bash, HTML/CSS

Frameworks & Libraries: FastAPI, Node.js, React, Express, JUnit

Databases & Tools: PostgreSQL, MySQL, InfluxDB, Git, Docker, Postman, Vim, Figma

Systems & Practices: Linux/Unix, CI/CD, Agile, DevOps, SDLC, Jira, AI